

A Survey of Graph Retrieval-Augmented Generation for Customized Large Language Models

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Abstract—Large language models (LLMs) have demonstrated remarkable capabilities in a wide range of tasks, yet their application to specialized domains remains challenging due to the need for deep expertise. Retrieval-augmented generation (RAG) has emerged as a promising solution to customize LLMs for professional fields by seamlessly integrating external knowledge bases, enabling real-time access to domain-specific expertise during inference. Despite its potential, traditional RAG systems, based on flat text retrieval, face three critical challenges: (i) complex query understanding in professional contexts, (ii) difficulties in knowledge integration across distributed sources, and (iii) system efficiency bottlenecks at scale. This survey presents a systematic analysis of Graph-based Retrieval-Augmented Generation (GraphRAG), a new paradigm that revolutionizes domain-specific LLM applications. GraphRAG addresses traditional RAG limitations through three key innovations: (i) graph-structured knowledge representation that explicitly captures entity relationships and domain hierarchies, (ii) efficient graph-based retrieval techniques that enable context-preserving knowledge retrieval with multihop reasoning ability, and (iii) structure-aware knowledge integration algorithms that leverage retrieved knowledge for accurate and logical coherent generation of LLMs. In this survey, we systematically analyze the technical foundations of GraphRAG and examine current implementations across various professional domains, identifying key technical challenges and promising research directions. All the related resources of GraphRAG, including research papers, open-source data, and projects, are collected for the community in <https://github.com/DEEP-PolyU/Awesome-GraphRAG>.

Index Terms—Retrieval-Augmented Generation, Knowledge Graphs, Graph-based Retrieval-Augmented Generation

I. INTRODUCTION

Large language models (LLMs), like the GPT series [1], have surprised the world with their remarkable ability across a wide range of tasks, achieving breakthroughs in text comprehension [2], question answering [3], and content generation [4]. Despite their effectiveness, LLMs are always criticized for their limited ability to handle knowledge-intensive tasks, especially when faced with questions requiring domain expertise [5]. Specifically, applying LLMs into specialized domains remains challenging for three fundamental reasons, including ❶ Knowledge limitations: LLMs’ pre-trained knowledge is broad but

shallow in specialized fields. Their training data primarily consists of general-domain content, leading to insufficient depth in professional domains and potential inconsistencies with current domain-specific standards and practices. ❷ Reasoning complexity: Specialized domains require precise, multi-step reasoning with domain-specific rules and constraints. LLMs often struggle to maintain logical consistency and professional accuracy throughout extended reasoning chains, particularly when dealing with technical constraints or domain-specific protocols. ❸ Context sensitivity: Professional fields often involve context-dependent interpretations where the same terms or concepts may have different meanings or implications based on specific circumstances. LLMs frequently fail to capture these nuanced contextual variations, leading to potential misinterpretations or inappropriate generalizations.

To adapt LLMs for specific or private domains, initial strategies involved **fine-tuning LLMs** with specialized datasets [6]. This method enhances performance by adding a limited number of parameters while fixing the parameters learned in the pre-training [7]. However, the significant distribution gap between the domain-specific dataset and the pre-training corpus makes it challenging for LLMs to integrate new knowledge without compromising their existing understanding [8]. A recent study by Google Research further highlighted the risks associated with using supervised fine-tuning to update knowledge, particularly in cases where new knowledge conflicts with pre-existing information; acquiring new knowledge through supervised fine-tuning can lead to the model generating new hallucinations and even experiencing severe catastrophic forgetting [9].

Retrieval-augmented generation (RAG) offers a promising solution to customize LLMs for specific domains [10], [11]. Rather than retraining LLMs to incorporate updates, RAG enhances these models by leveraging external knowledge from text corpora without modifying their architecture or parameters. This approach enables LLMs to generate responses by leveraging not only their pre-trained knowledge but also real-time retrieved domain-specific information, thereby providing more accurate and reliable answers. The naive RAG systems operate through three key steps: knowledge preparation, retrieval, and integration. During knowledge preparation, the external textual corpus is divided into manageable textual chunks and converted into vector representations for efficient indexing. In the retrieval stage, when a user submits a query, the system searches for relevant chunks using keyword matching or vector similarity measures. The integration stage then combines these retrieved chunks with the original query to generate more informed responses. Recently, some advanced RAG systems have evolved

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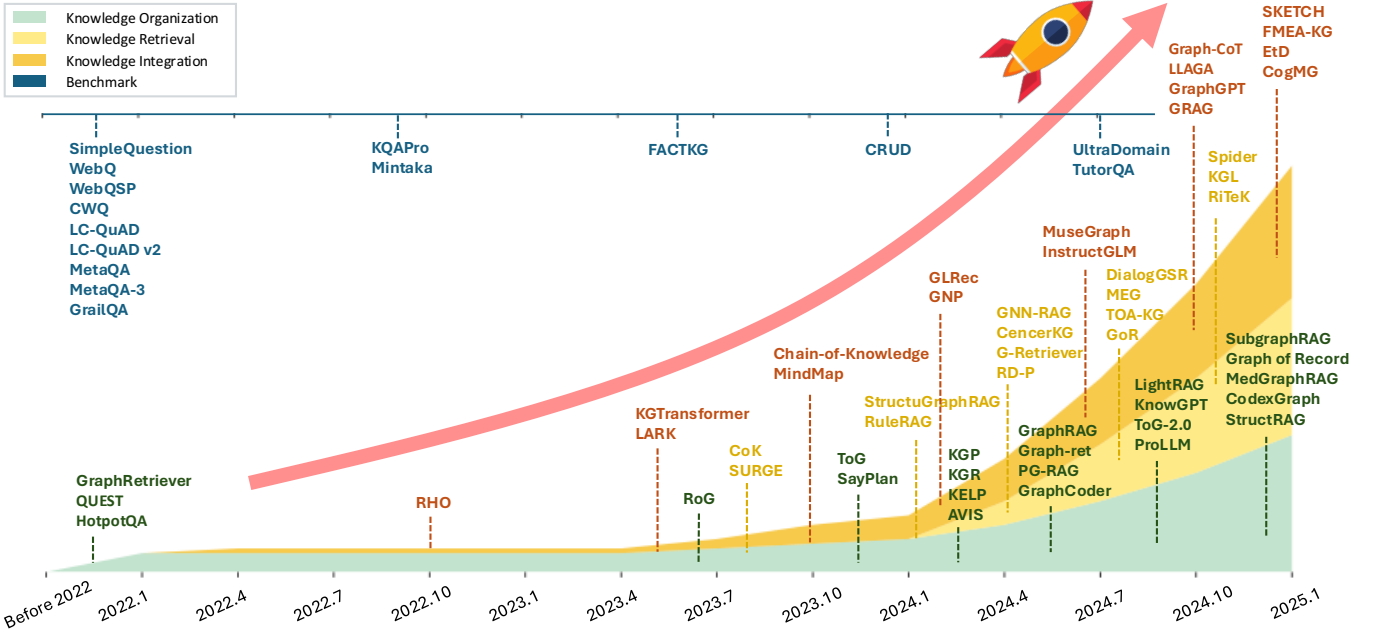


Fig. 1: The development trends in the field of GraphRAG with representative works.

beyond simple text chunk retrieval to offer more sophisticated knowledge augmentation approaches. These include hierarchical RAG that preserves document structure through multi-level retrieval [12], [13], re-ranking systems that implement two-stage retrieval for higher recall and precision [14], [15], self-querying RAG that automatically decomposes complex queries [16], and adaptive RAG that dynamically adjusts retrieval strategies based on query types [17], [18]. These strategies advance naive RAG systems by improving context awareness, retrieval accuracy, and handling complex queries.

The emergence of RAG has offered a promising approach for customizing LLMs with domain-specific knowledge. However, despite its potential, RAG faces several critical limitations that impact its effectiveness in practical applications. These limitations can be broadly categorized into four main challenges that significantly affect the performance and utility of RAG-enhanced LLMs. The primary challenge lies in **complex query understanding**. Specialized domains often involve intricate terminology and industry-specific jargon that requires precise interpretation [19]. User queries in these areas typically contain numerous technical terms and industry-specific expressions, with solutions often requiring reasoning across multiple related concepts. Traditional RAG approaches, which rely on simple keyword matching and vector similarity techniques, are inadequate for capturing the deep semantic nuances and multi-step reasoning processes necessary for accurate and comprehensive [20]. For instance, when queried about the connection between concept A and concept D, these systems typically retrieve only directly related information, missing crucial intermediate concepts like B and C that could bridge the relationship. This narrow retrieval scope limits RAG’s ability to a broad contextual understanding and complex reasoning.

Another key challenge involves **integrating domain knowledge from distributed sources**. Domain knowledge is usually collected from different sources, such as textbooks, research papers, industry reports, technical manuals, and maintenance

logs. These textual documents may have varying levels of quality, accuracy, and completeness. The retrieved knowledge is often flattened, extensive, and intricate, while domain concepts are typically scattered across multiple documents without clear hierarchical relationships between different concepts [5], [21], [22]. Although RAG systems attempt to manage this complexity by dividing documents into smaller chunks for effective and efficient indexing, this approach inadvertently sacrifices crucial contextual information, significantly compromising retrieval accuracy and contextual comprehension. This limitation hampers the ability to establish robust connections between related knowledge points, leading to fragmented understanding and reduced efficacy in leveraging domain-specific expertise.

The third limitation stems from the **inherent constraints of LLMs**. While RAG systems can retrieve relevant information from vast knowledge bases, the LLM’s ability to process this information is constrained by its fixed context window (typically 2K-32K tokens) [1], [23]. Long-range dependencies in complex documents cannot be fully captured, as the content exceeding the context window must be truncated or summarized, disrupting natural semantic units and logical flow. The challenge of maintaining coherence across extensive knowledge contexts becomes increasingly problematic in professional domains, as critical information may be lost during context window truncation. This fundamental limitation directly impacts the system’s ability to process and synthesize comprehensive information from large-scale knowledge bases.

The last challenge relates to system **efficiency and scalability**. The entire RAG pipeline - from initial corpus preprocessing and indexing to real-time retrieval and generation - faces significant efficiency bottlenecks [24], [25]. The external knowledge base contains a lot of domain-irrelevant information, while domain-specific terminologies are always sparsely distributed over these documents. RAG systems can be computationally expensive and time-consuming [24], especially when dealing with large-scale knowledge sources, as the model needs to